

Step-1

- > Create New Project
- > select Java With Ant
- > Select Java Application
- > Write the Project Name
- > Finish

Step-2

- > Open Project Window
- > Right click on Project name > New > Java Interface > Name the interface (Calculator) > Finish
- > Write code in Calculator interface

[Code]

```
import java.rmi.Remote;  
import java.rmi.RemoteException;  
public interface Calculator extends Remote {  
    int add(int x, int y) throws RemoteException;  
    int subtract(int x, int y) throws RemoteException;  
}
```

Step-3

- > Open Project Window
- > Right click on Project name > New > Java Class > Name the Class (CalculatorImpl) > Finish
- > Write code in CalculatorImpl class

[Code]

```
import java.rmi.RemoteException;  
import java.rmi.server.UnicastRemoteObject;  
public class CalculatorImpl extends UnicastRemoteObject implements  
    Calculator {  
    public CalculatorImpl() throws RemoteException {  
        super();  
    }  
    public int add(int x, int y) throws RemoteException {  
        return x + y;  
    }  
    public int subtract(int x, int y) throws RemoteException {
```

```
return x - y;
}
}
```

Step-4

--> Open Project Window
--> Right click on Project name > New > Java Class > Name the Class (CalculatorServer) > Finish
--> Write code in CalculatorServer class

[Code]

```
import java.rmi.RemoteException;
import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
public class CalculatorServer {
    public static void main(String[] args) {
        try {
            // Create an instance of the remote object
            Calculator calculator = new CalculatorImpl();
            // Get a reference to the RMI registry
            Registry registry = LocateRegistry.createRegistry(1099);
            // Bind the remote object to the registry
            registry.rebind("Calculator", calculator);
            System.out.println("CalculatorServer ready.");
        } catch (RemoteException e) {
            System.err.println("CalculatorServer exception:");
        }
    }
}
```

Step-4

--> Open Project Window
--> Right click on Project name > New > Java Class > Name the Class (CalculatorClient) > Finish
--> Write code in CalculatorClient class

[Code]

```
import java.rmi.NotBoundException;
import java.rmi.RemoteException;
import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
public class CalculatorClient {
    public static void main(String[] args) {
        try {
            // Get a reference to the RMI registry
            Registry registry =
                LocateRegistry.getRegistry("localhost");
            // Look up the remote object from the registry
            Calculator calculator = (Calculator)
                registry.lookup("Calculator");
            // Invoke the methods on the remote object
            int result = calculator.add(1, 2);
            System.out.println("1 + 2 = " + result);
            result = calculator.subtract(5, 3);
            System.out.println("5 - 3 = " + result);
        } catch (NotBoundException | RemoteException e) {
            System.err.println("CalculatorClient exception:");
        }
    }
}
```

Step-4

--> Clean And build all Four files (Calculator, CalculatorImpl, CalculatorServer, CalculatorClient)

Step-5

--> Go to project window < Select project < Libraries < Right click on libraries < Add Jar files > ()

Step-6

--> Run CalculatorServer File

Step-7

--> Run CalculatorClient file